LUNGRAY: MULTI-TASK CLINICAL CADX AND TRUSTWORTHY WSOL GROUNDING PLATFORM

CAPSTONE PROJECT PROPOSAL (REPORT #1) — COURSE CODE: DSP391

Specialization: Artificial Intelligence & Data Science | Department of Computer Science
FPT University, Ho Chi Minh City Campus — November 2026

Project Team (Group 11): Tan Phat & Duc Manh
Academic Supervisor: Teacher Nguyen Trong Tai

I. PROJECT INTRODUCTION

1. Overview

1.1 Project Information

The operational metadata, team governance, and institutional stakeholders for the LungRay capstone initiative are systematically documented in Table 1.1.

Table 1.1: Project Information and Organizational Metadata.

Attribute	Detailed Specification
Project Name	LungRay: Multi-Task Clinical CADx and Trustworthy WSOL Grounding Platform
Project Code	DSP391
Team Members	Tan Phat (Lead AI Architect & Systems Engineer) Manh (AI Research & Data Operations Engineer)
Key Stakeholders	Radiologists (Triaging & Second Reader), Emergency Clinicians, Capstone Academic Committee
Core Architecture	Multi-Task 3-Head CNN (Shared DenseNet-121 Encoder, 512×512 Input Resolution)
Diagnostic Scope	16 Cardiopulmonary Conditions (14 NIH Thoracic Pathologies + Tuberculosis + COVID-19)
Safety Systems	Helmholtz Free Energy Out-of-Distribution Lockout (Tau_E = 9.0421) & Multi-Scale Grad-CAM++ IoU

1.2 Project Overview

Chest radiography (Chest X-ray, CXR) constitutes the single most frequently performed projectional imaging examination in modern clinical medicine, representing over 40% of all diagnostic imaging workloads worldwide. Radiologists routinely inspect projectional radiographs to detect pulmonary infections, thoracic masses, cardiac enlargement, pleural effusion, and chronic parenchymal disorders. However, healthcare institutions globally confront a dual crisis: exponential growth in scan volumes contrasted against an acute shortage of certified radiologists. This disparity induces substantial cognitive fatigue, prolonged diagnostic turnaround times (24 to 72 hours in resource-constrained environments), and diagnostic discrepancy rates between 10% and 30% in high-pressure emergency triaging.

The LungRay initiative establishes an enterprise-grade Computer-Aided Diagnosis (CADx) and Clinical Decision Support (CDS) platform specifically designed to overcome these institutional bottlenecks. Moving beyond conventional, siloed single-disease models, LungRay formulates a unified Multi-Task 3-Head Convolutional Neural Network driven by a shared DenseNet-121 visual encoder. The platform executes simultaneous, holistic screening across sixteen cardiopulmonary pathologies while enforcing Weakly-Supervised Object Localization (WSOL) via Multi-Scale Grad-CAM++ to render decisions visually and mathematically interpretable.

2. Project Background

Projectional chest radiography provides a high-resolution, low-cost anatomical survey of thoracic structures, including the pulmonary parenchyma, pleural spaces, mediastinum, cardiac silhouette, and bony thorax. Interpreting planar CXRs represents an intricate perceptual and cognitive challenge. Cardiopulmonary pathologies manifest as subtle gradations in radiopacity, overlapping soft-tissue shadows, and ambiguous anatomical margins. For instance, early-stage solitary pulmonary nodules or retrocardiac consolidations frequently obscure within normal vascular markings or diaphragmatic contours.

In low- and middle-income healthcare systems, regional referral clinics frequently operate without dedicated on-site thoracic radiologists. General practitioners must review complex radiographs under extreme time constraints. In infectious disease epidemics—such as active pulmonary tuberculosis surges or viral pneumonia outbreaks—delayed identification of positive cases accelerates nosocomial transmission and compromises patient prognoses.

While deep learning has demonstrated expert-level accuracy on curated datasets, clinical translation has been severely obstructed by three systemic deficiencies in prior medical AI systems:

- Task Siloization and Memory Redundancy: The vast majority of published models address only a solitary diagnostic target (e.g., exclusively detecting COVID-19 or solely binary pneumonia). Deploying multiple disjoint single-task models requires loading dozens of discrete neural networks into memory, resulting in unsustainable GPU VRAM overhead and fragmented inference pipelines.
- Extreme Positive-Negative Class Imbalance: Within multi-label thoracic radiography cohorts (such as the NIH ChestX-ray14 dataset), positive prevalence for critical conditions like Hernia (0.2%) or Pneumothorax (4.7%) is heavily suppressed by dominant negative instances (over 95% to 99% negative per pathology). Traditional Binary Cross-Entropy loss functions collapse toward the majority negative class, failing to optimize discriminative boundaries for critical minority findings.
- Confounder Memorization (Shortcut Learning): Deep convolutional networks routinely learn non-physiological shortcuts (such as hospital-specific metal orientation markers 'R'/'L', radiopaque wires, and detector border artifacts) rather than true pathology, resulting in failure when tested across external hospitals.

3. Project Objectives

The overarching engineering objective of the LungRay platform is to develop, validate, and deploy a trustworthy multi-task deep learning system that delivers rapid, calibrated, and interpretable chest radiograph triaging without violating clinical safety constraints. The concrete measurable objectives comprise:

- **Unified Multi-Task Architecture:** Construct MultiTaskChestXrayNet operating at 512×512 resolution, isolating gradients across 3 heads via parameter-free LayerNorm feature projection.
- **Zero Patient-Level Data Leakage:** Enforce strict patient-wise partitioning ensuring zero overlapping Patient_ID instances across train/val/test splits (70/10/20 on NIH ChestX-ray14), preventing artificial performance inflation.
- **Multi-Task Batch Equilibrium:** Formulate WeightedMultiTaskBatchSampler maintaining an exact ratio of 60% NIH : 25% COVIDx : 15% TB per mini-batch, eliminating gradient starvation across heterogeneous tasks.
- **Imbalance Mitigation via Asymmetric Loss:** Implement Asymmetric Loss (ASL) with probability margin m = 0.05 to elevate minority-class multi-label ROC-AUC above 0.82 on Head A without suppressing minority classes.
- **Trustworthy WSOL Localization:** Deploy Multi-Scale Grad-CAM++ achieving Mean Box-IoU >= 0.35 against expert radiologist bounding boxes (VinDr-CXR and RSNA) without box supervision during training.
- **Shortcut Learning Audit & Defeat:** Enforce automated dilated lung masking, ensuring that >= 80% of gradient activation energy resides within anatomical lung fields on COVID-19 positive radiographs.
- **Clinical Fail-Safe Layer:** Deploy Helmholtz Free Energy Out-of-Distribution screening to identify and reject corrupted or non-CXR inputs prior to classification.
- **Sub-Two-Second PACS Serving:** Containerize the platform into an asynchronous FastAPI backend delivering inference within <= 1.8 seconds, paired with a Next.js 14 Medical PACS Darkroom Viewer (#070B12).

4. Problem Statement

Formally, let an input frontal chest radiograph be formalized as an image tensor X in R^(3 × 512 × 512). The multi-task CADx system must map X to a compound diagnostic prediction vector Y_hat = {y_hat_A, y_hat_B, y_hat_C} parameterized by a shared convolutional encoder f_theta(X):

- $y_hat_A = sigma\left(g_phi_A\left(f_theta(X)\right)\right) in [0,1]^{14}$ (Bernoulli probabilities for 14 NIH thoracic findings)
- $y_hat_B = sigma\left(g_phi_B\left(f_theta(X)\right)\right) in [0,1]$ (Scalar probability of active pulmonary tuberculosis)
- $y_hat_C = Softmax\left(g_phi_C\left(f_theta(X)\right)\right) in [0,1]^3$ (Probabilities for COVID-19 differential triage)

The fundamental machine learning challenges in solving this problem simultaneously include resolving disjoint dataset label spaces, overcoming the 98.6% negative label ratio in multi-label loss gradients, extracting multi-focal lesion bounding boxes without spatial supervision, and preserving real-time clinical latency.

5. Significance of the Project

- **Clinical Impact & Patient Outcomes:** Functions as an automated intelligent second-reader. In emergency departments, it automatically triages incoming CXRs, flagging life-threatening conditions (such as Tension Pneumothorax, Massive Pleural Effusion, or Severe Bilateral Consolidation) for immediate, prioritized radiologist review within two minutes of image capture.

- **Academic & Methodological Novelty:** The project provides empirical proof that a single shared DenseNet-121 encoder can achieve discriminative performance matching or exceeding three independent single-task networks while slashing GPU memory requirements by 66%. Furthermore, our quantitative Bounding Box IoU benchmarking establishes an objective standard over purely qualitative heatmap inspection.
- **Economic & Healthcare Resource Optimization:** By containerizing the entire inference pipeline into a lightweight Docker microservice with dynamic CPU/GPU fallback, LungRay can be deployed on low-cost edge servers in rural provincial hospitals without requiring continuous internet connectivity or multi-thousand-dollar GPU clusters.

6. Project Scope & Limitations

To establish rigorous clinical safety and prevent inappropriate system usage, explicit boundaries are defined:

- **In-Scope Capabilities:** Adult frontal projection chest radiographs acquired in Posteroanterior (PA) or Anteroposterior (AP) views; automated scoring across 16 specified thoracic conditions; interactive Grad-CAM++ overlays; Free Energy OOD safety screening; responsive Next.js darkroom viewer.
- **Pediatric Radiographs Excluded:** Anatomical proportions, ossification centers, and normal cardiothoracic ratios in neonates and young children differ significantly from adults; pediatric cohorts are strictly excluded from the operational scope.
- **Lateral Projection CXRs Excluded:** Only frontal projection radiographs are supported; lateral chest views are beyond current processing capabilities.
- **Non-Autonomous Classification:** LungRay is explicitly designed as an assistive second-reader (Class IIa Medical Software). It does not issue autonomous definitive clinical diagnoses or prescribe therapeutic regimens. All outputs require validation by a licensed medical practitioner.

7. Preliminary Research & Proposed Technical Approach

To ensure clinical generalization and eliminate institutional bias, the platform curates and standardizes six internationally recognized open-access medical imaging cohorts, encompassing over 142,000 chest radiographs (Table 1.2).

Table 1.2: Standardized Clinical Radiography Datasets in LungRay Platform.

Dataset Identifier	Clinical Scope	Cohort Size	Primary Function in Project
NIH ChestX-ray14	14 Thoracic Pathologies	112,120 CXRs	Head A Multi-Label Training & Testing
TB Chest Radiography	Pulmonary Tuberculosis	7,000 CXRs	Head B Tuberculosis Screening
COVIDx CXR-4	COVID-19 / Pneumonia / Normal	16,400+ CXRs	Head C COVID-19 Differential Triage
VinDr-CXR	Vietnamese Thoracic Cohort	3,000 CXRs	Quantitative WSOL IoU External Benchmark
RSNA Pneumonia	Pneumonia Consolidations	6,000+ CXRs	Quantitative WSOL IoU External Benchmark

Dataset Identifier	Clinical Scope	Cohort Size	Primary Function in Project
Stanford CheXpert	External Multi-Label Cohort	1,000 CXRs	Out-of-Distribution Generalization Audit

8. Project Execution Plan & Milestones

The project is structured across an accelerated 9-week execution lifecycle, organized into agile sprint deliverables and verified through six formal Quality Delivery Gates (Table 1.3).

Table 1.3: Nine-Week Project Execution Roadmap and Quality Delivery Gates.

Timeline	Phase Focus	Engineering Deliverables	Quality Delivery Gate
Week 1	Foundations & Setup	Literature review, taxonomy definition, Git repo and PyTorch environment setup.	Workspace Ready
Week 2	Data Ingestion & Splits	Patient-wise split (Zero Leakage), 512×512 letterbox, multi-task sampler 60:25:15.	Gate 1: Data Integrity
Week 3	Multi-Task Baseline	MultiTaskChestXrayNet architecture, masked backpropagation, baseline checkpoint.	Gate 2: Baseline Model
Week 4	Loss Optimization	Asymmetric Loss (ASL) and dynamic Focal Loss, Youden's J threshold calibration.	Gate 3: Loss Stability
Week 5	XAI & Shortcut Audit	Grad-CAM++, VinDr & RSNA Box-IoU benchmarking, in-lung activation audit (>= 80%).	Gate 4: WSOL Grounding
Week 6	Full-Stack PACS Serving	FastAPI asynchronous microservice (<= 1.8s) and Next.js 14 PACS viewer integration.	PACS Integration
Week 7	Clinical Fail-Safe Layer	Helmholtz Free Energy OOD lockout implementation and structured ICD-10 reporting.	Gate 5: Clinical Safety
Week 8	Regression Testing	138 automated unit tests PASS (verify.ps1), container packaging, latency optimization.	Regression Closure
Week 9	Final Thesis & Defense	Comprehensive thesis report, 40-slide defense presentation deck, video walkthrough.	Gate 6: Final Defense

II. PROPOSAL SUMMARY & SIGN-OFF

The LungRay platform addresses an urgent, high-impact clinical challenge by coupling state-of-the-art multi-task deep learning with trustworthy explainability and out-of-distribution clinical safety guardrails. Group 11 is committed to executing this project under rigorous scientific integrity, delivering a production-grade medical artificial intelligence platform ready for clinical decision support.